Coverage, Context, Convention: Restoring Vowel Length in Old English

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Abstract

Many editions of Old English mark long vowels with macrons, but the manuscripts do not mark length systematically, so bare text merges *god* 'God' and *gōd* 'good'. We restore the marks by choosing, for each bare word, among the length patterns that candidate sources list, and we ask what limits the result: missing candidates, too little context, or the editor's convention. We build a length-marked corpus of nine editions (161,062 tokens), to our knowledge the only one besides Tauber's *Beowulf*, and release it with Tauber's text. Missing candidates are the largest limit: with WikiPron candidates, 64.8% of a frequency prior's prose errors and 73.4% of its verse errors have no correct candidate. Context helps a group of five closed-class words, chiefly *þe/þē*, and barely changes the rest: on prose, a ByT5 scorer that sees two words on each side raises accuracy on the five from 62.66% to 75.40%, and a post hoc logistic model with word-specific collocations does about as well. On ambiguous verse tokens context adds little, partly because of one unseen name (post hoc). Convention can outweigh context: swapping the gold between two *Beowulf* editions moves the prior's score on the five words by 11.81 points, against a context gain of +7.39 on the same tokens with Klaeber as gold. CharsiuG2P's published Old English word error rate (0.208) partly reflects a 20-step decoding limit (0.188 with 50 steps).

Code and datasets are available at:
- https://github.com/Ace3Z/coverage-context-convention
- https://huggingface.co/datasets/Ace3Z/old-english-vowel-length

1 Introduction

Many editions of Old English mark long vowels with a macron. The manuscripts do not mark length systematically, though some Late Old English scribes marked it with accents [18]. Bare text merges *god* 'God' and *gōd* 'good', or the relative particle *þe* and the pronoun *þē* 'thee', so restored marks could help readers of texts transcribed without them. Restoring them means choosing, for each word, among the length patterns that candidate sources list: lexical disambiguation as in accent restoration [35] and homograph disambiguation in text-to-speech [12], here for a historical language with little data. The ambiguity is lexical (class A below, *god*), grammatical (class B, *þe*) or editorial (class C). We predict an editor's marking convention, not pronunciation.

To our knowledge, no system has restored Old English vowel length in context. We measure three limits on the same two test texts. *Coverage* (Section 5.1) asks whether the right candidate exists, and with WikiPron candidates, for most errors of a frequency prior (which picks each word's most frequent pattern), it does not. *Context* (Section 5.2) asks whether neighbouring words can pick it out, and they help mainly class B, chiefly *þe/þē*. *Convention* (Section 5.3) asks whose marking the gold follows, and on class B the choice of *Beowulf* edition can move the prior more than context adds. We contribute a length-marked corpus of nine editions (Section 2, to our knowledge the only one besides Tauber's *Beowulf* [26]), an evaluation that separates the three limits, a reanalysis of CharsiuG2P [37] (Section 5.4) and tested code whose one script recomputes every result here from the released data and model predictions (MIT licence).

Related work. Yarowsky [35] restored Spanish and French accents with decision lists over collocations. He calls accented text a fully objective standard, yet found his corpus wrong on about half of the disputed tokens for two ambiguities, and Old English editions also differ by convention. Latin [31] and Ancient Greek [28] macronisers use taggers, lemmatisers and morphological analysers or large pretraining [7]. Winge [31] already bounded his Latin system with a lexicon oracle and traced the largest group of its disagreements with a hand-marked text to marking conventions. His tagging and lemmatisation add under a point of vowel accuracy, close to what context adds here over all tokens. Gershuni and Pinter [11] restore Hebrew diacritics with a character tagger instead of candidates, and Esmail et al. [10] measure how much lookahead helps Arabic diacritisation. We test lookahead with a model that sees only preceding words. Plank [22] argues that one gold label hides annotator variation, which here includes editorial convention.

2 Data set

Table 1 lists the corpus: nine editions printed from 1880 to 1921 with 161,062 tokens in 7,889 sentences, plus Tauber's *Beowulf*. We took Project Gutenberg and Wikisource transcriptions [2, 3, 4, 6, 13, 14, 25, 33, 38] and cleaned them with character mappings, each counted in a manifest (some files cannot be rebuilt exactly from our repository). The Hugging Face release (CC BY-SA 4.0) keeps each sentence's work, editor, year, genre and section, per-token length patterns and a leakage flag. We built the corpus because word lists lack context, treebanks mark length at most on lemmas (UD OEDT [17]) or, as far as we know, not at all (YCOE [27]), and Tauber's *Beowulf* is a single poem. With it, our corpus holds three editions of *Beowulf* to compare.

Splits are by work, so no test work is seen in training. Training (69,087 tokens, without Zupitza, Section 4) is mostly prose and 85.6% Bright's. The development text, John, shares a Gospel series with the Matthew training text and is easier than the tests: 6.36% of its tokens are unseen in training, against 17.94% (prose test) and 31.53% (verse test). A word 6-gram audit removes 34 John sentences and 1 *Soliloquies* sentence from scoring, and the longest word run that *Beowulf* shares with training has 5 words. The verse gold is Brunetti's tokenisation of *Beowulf* with the lengths of the Perseus text of Klaeber's edition [15], transferred by Tauber.

3 Preprocessing

Preprocessing normalises Unicode (NFC), turns circumflexes into macrons, drops other diacritics and English and Latin lines (by their function words), splits sentences at end punctuation and

Table 1: The corpus. PG: Project Gutenberg, WS: Wikisource. % long: share of vowel letters marked long.

Edition	Editor, printing	Source	Genre	Tokens	% long	Role
Ælfric's *Grammar*	Zupitza 1880	PG	prose	23,940	23.58	ablation
Anglo-Saxon Reader	Bright 1913	WS	mixed	38,484	23.91	train
Gospel of Matthew	Bright, 1910 printing	WS	prose	20,648	25.86	train
Anglo-Saxon Primer	Sweet 1905	PG	prose	9,003	24.20	train
Physiologus	Cook 1921	PG	verse	952	20.57	train
Gospel of John	Bright 1904	WS	prose	17,310	26.14	dev
Alfred's *Soliloquies*	Hargrove 1902	WS	prose	15,475	26.11	prose test
Beowulf	Harrison & Sharp 1894[a]	PG	verse	17,682	21.24	agreement
Beowulf	Wyatt 1894[b]	WS	verse	17,568	20.74	agreement
Beowulf	Klaeber 1922	Tauber	verse	17,361	21.22	verse test

[a]Project Gutenberg revision of the 4th edition (first edition 1883). [b]Its *Finnsburh Fragment* lacks lines 1 to 18 in our file (187 tokens, against 297 in Harrison and Sharp). Scored tokens: dev 16,649, prose 15,426, verse 17,361. Hugging Face release: 8,650 sentences, 178,423 tokens.

takes letter runs, joined across hyphens and apostrophes, as tokens. A token's *key* is its lowercased form without macrons and with ð written þ, and its *pattern* has one digit per vowel letter, 1 for a macron: Harrison and Sharp's *geārdagum* (0100) and Klaeber's *gēardagum* (1000) share the key *geardagum* and differ only in which digraph letter bears the macron.

A key's candidates are its training patterns plus those of every listed form with that key. Our main list is WikiPron [16], whose IPA we convert to spellings by aligning vowel units with IPA nuclei (97.6% of rows align). A coverage comparison adds the dictionaries of our proposal, Clark Hall [23], Wiktionary via kaikki.org [36] and Bosworth-Toller [29], though WikiPron is itself mined from Wiktionary. A token is *lexicon-ambiguous* when its key has two or more candidates, as for 37.8% of prose and 25.4% of verse tokens with WikiPron. As adding sources enlarges this set, we compare sources on all tokens. The 60 most frequent ambiguous keys (ranked before we dropped Zupitza, with test tokens counted but not their gold) carry a class: 13 lexical (A: *god/gōd*), 5 grammatical (B: *þe, ne, ge, gyt, þis*) and 42 editorial (C: *se/sē, he/hē*), covering most ambiguous tokens.

Labels. The classes, the types of 400 sampled errors and a sample of 100 editor disagreements were labelled by a large language model, Claude Opus 5.5 [1], with written guidelines. For the classes and the error types, two runs of the model labelled every item separately (Cohen's κ 0.927 and 0.975 [5]), and a third run settled their disagreements. A single run labelled the disagreement sample. A blind review by further runs of the same model changed 7 of the 560 labels. The labels were then manually checked, and none was changed. As both annotators are runs of one model, κ measures that model's consistency rather than agreement between people, and as the manual check was not blind to the model's labels, we lack the comparison with independent human labels that Pangakis et al. [19] call for.

4 Model training

Baselines. All our systems are supervised, choose among the same candidates and leave a word without candidates unmarked. The frequency prior, fixed in our proposal after Yarowsky [35], picks a word's most frequent training pattern and, for an unseen word, the pattern most sources list, with ties going to fewer long vowels. Logistic classifiers (scikit-learn [21], untuned, three seeds) see sparse indicators of the word and its first and last two letters (*word*), plus the two preceding words (*left*), plus the two following words (*bilateral*). These context features are not conjoined with the word, so 'previous word *tō*' shifts every word's scores alike. After the ByT5 results we added a post hoc *conjoined* variant with the word-by-neighbour features of earlier work [12, 35].

ByT5. As our main model we fine-tune ByT5-small [34] with PyTorch [20] and Transformers [32] to score candidates. ByT5 reads bytes, so thorn, eth and ash need no vocabulary. It is also CharsiuG2P's backbone, and scoring candidates guarantees a valid output. The input is the bare word, alone or in context with the target marked `[[ ]]`, the output is its pattern string, and a candidate's score is the summed log-probability of its bytes, free of length bias since candidates

are equally long. Arms differ only in the input: *word*, *left* (all preceding words), *sentence*, *local2* and *local5* (two or five words on each side), all trained on the same 25,949 ambiguous training tokens as the logistic models. We report the full input cap (2,560 bytes), as 128- and 256-byte caps move mean ambiguous-token accuracy by under two points.

Protocol. We chose the setting (learning rate $3 \cdot 10^{-4}$, 5 epochs, no Zupitza) by mean ambiguous-token development accuracy (two seeds per setting, 24 sentence-arm runs) and reused it for every arm. Dropping Zupitza's *Grammar*, which marks *þes* and *gyt* short, lifts the prior and the bilateral model on ambiguous development tokens by about half a point each. ByT5 gains under half a point in four of the five settings run both ways and changes by -7.40 in the least trained (we leave out a sixth, where one run collapsed onto the candidate with most long vowels). The price, measured afterwards, is coverage: only Zupitza writes *forþām*, *swāswā* and *forþī* as one word, so 153 prose test tokens become unknown. Over all prose tokens, dropping Zupitza costs the prior 0.88 points and the ByT5 sentence arm 1.18. The final grid has 39 runs. About fifty exploratory runs were scored on the test sets before the development set existed, and the selection runs also logged test scores. The setting was chosen on development scores only.

Changes from the proposal. The corpus grew to nine editions. A ByT5 encoder-decoder replaced the planned encoder classifier. WikiPron superseded the proposed dictionaries as the main candidate source. We added the logistic baselines and the local2 and local5 arms, and chose local2 as the headline arm after the test results. The class B focus and the gold swap of Section 5.3 were not planned. An LLM labelled what we had planned to label ourselves.

5 Evaluation

Our main finding: missing candidates cost more than context adds on both test texts, and on *Beowulf* class B the choice of gold edition can move the prior's score more than context adds. We score token accuracy, as a reader needs the whole word: a token counts only if every vowel has the gold length. On all tokens, marking nothing scores 57.58% (prose) and 59.11% (verse), and the WikiPron prior 87.85% and 89.40%. Differences carry 95% intervals from a paired bootstrap over sentences [9], whose tokens are not independent (1,000 draws, correctness averaged over runs, runs held fixed). Intervals and p-values are not corrected for multiple comparisons, and we claim no difference between ByT5 arms below about one point.

5.1 Coverage

Most of the prior's errors are beyond any chooser: with WikiPron, 64.8% of its prose errors and 73.4% of its verse errors have no correct candidate, mostly because the word is unknown. Our proposal had assumed that every error is a selection error. Both tests have about the same achievable ceiling, a perfect chooser's score (Figure 1, about 92%), and a better chooser could gain at most 4.28 points over the prior on prose and 2.82 on verse. Dictionaries change the prior only for words unseen in training, as a dictionary-only pattern has no training count: on unseen verse tokens all four sources lift it from 52.89% to 85.02%, and 94% of this gain (10.13 points over all verse tokens) falls on *Beowulf*-only word types, so it may not hold for other verse.

5.2 Context

Context helps class B and barely moves classes A and C. On prose, local2 beats the word arm by $+2.39$ $[+1.96, +2.88]$ points on lexicon-ambiguous tokens (Table 2), almost all of it in class B, where accuracy rises from 62.66% (the word arm equals the prior there) to 75.40%. Every context arm gains there, and the left arm, which sees no following word, gains at least as much as local2. Esmail et al. [10] find one word of lookahead dramatic for Arabic diacritisation, yet here the left arm, with none, matches local2 on all ambiguous prose tokens. Most of the class B gain comes from telling the relative *þe* from the pronoun *þē* 'thee' (57.7% to 76.5% of 567 tokens). In *Ic clypie tō þē, Drihten, þū þe æall geworhtest* 'I call to thee, Lord, thou who madest all', the prior, the ByT5 word runs and the logistic word, left and bilateral runs mark both *þe* short, while every local2 run and every conjoined run gets the pronoun long and the relative short.

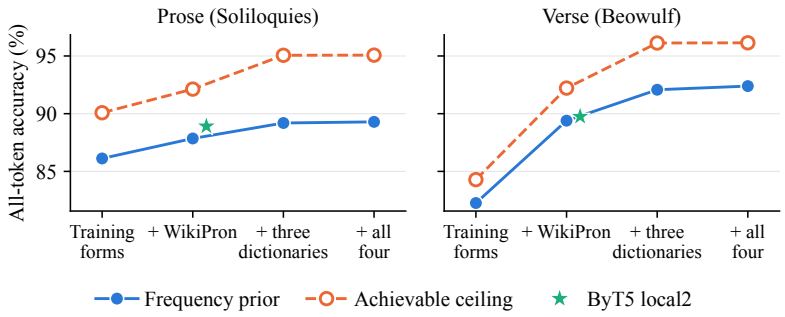


Figure 1: Coverage. All-token accuracy of the frequency prior and the achievable ceiling (gold among the candidates, or no candidate and an all-short gold). '+ three dictionaries' leaves out WikiPron (derived from Wiktionary), and '+ all four' adds it. Star: ByT5 local2 with WikiPron candidates, mean of three runs.

Table 2: Context. Accuracy (%) at the full byte cap, mean over three runs, ± SD for lexicon-ambiguous (Lex.-amb.) tokens and class B. Classes A, B, C: LLM labels, manually checked. †Post hoc diagnostic.

	Prose (*Soliloquies*)				Verse (*Beowulf*)			
	Lex.-amb.	A	B	C	Lex.-amb.	A	B	C
n	5,834	663	1,023	3,275	4,402	387	420	2,645
Frequency prior	88.46	97.44	62.66	95.85	88.85	89.41	71.19	95.80
Logistic bilateral	89.67±0.05	96.88	67.64±0.59	95.85	88.69±0.08	89.41	73.33±0.24	95.80
Logistic conjoined†	91.26±0.15	97.44	76.02±0.96	95.85	89.15±0.11	89.41	77.38±0.95	95.80
ByT5 word	88.88±0.08	97.44	62.66±0.00	95.85	89.97±0.03	89.41	71.19±0.00	95.80
ByT5 left	91.33±0.36	97.34	76.87±1.98	95.85	89.54±0.06	89.49	76.51±0.96	95.83
ByT5 sentence	90.34±1.26	97.49	70.90±7.09	95.85	89.29±0.13	89.41	75.08±3.42	95.90
ByT5 local2	91.27±0.52	97.13	75.40±3.32	95.88	90.26±0.44	89.41	78.41±2.38	95.80
ByT5 local5	91.85±0.76	97.44	79.05±3.87	95.86	89.75±0.16	89.41	77.62±2.65	95.80

On verse, local2 also gains on class B, again mostly from *þe*, but class B is only a tenth of the ambiguous verse tokens, and on all of them no context arm clearly beats the word arm (local2 $+0.29$ $[-0.11, +0.64]$). One name pulls the context arms down further. *Gēata* 'of the Geats' (39 tokens, all long) is unseen in training, and WikiPron also lists *geata* 'of gates'. The prior falls back to fewer long vowels and misses every token, and the word arm gets all of them in every run, while most context runs get few or none. Without *Gēata* (post hoc), local2 gains $+0.78$ $[+0.43, +1.12]$ points over the word arm, a small gain below our one-point margin. Over all tokens, context adds less than better coverage. Local2 adds 1.06 points to the WikiPron prior on prose and 0.36 on verse, while adding the three other dictionaries to WikiPron raises the prior by 1.44 and 3.00.

The bilateral logistic model gains less on prose class B (Table 2), as its features cannot express that, for example, a preceding *tō* matters for *þe* alone. The post hoc conjoined variant can, like the per-homograph classifiers of Gorman et al. [12], and on prose class B we cannot tell it from local2 (local2 minus conjoined: -0.62 $[-2.31, +1.05]$). Training variance limits the table. Local inputs never reach a cap, so each seed's runs at the three caps repeat one training: the local2 repeats agree exactly, yet the local5 repeats differ by up to 6.84 points on prose class B. Seeds [8] and tiny nondeterminism [24] move training results. We set no deterministic mode and did not test the cause, and three runs per arm cannot rank the context arms.

5.3 Convention

On class B, the choice of gold edition can move the prior's score more than context adds. Table 3 compares the three *Beowulf* editions on the tokens each pair aligns (identical spelling runs, over 93% of each text). Our Harrison and Sharp file, a Project Gutenberg revision that modernised some quantities (conjunctive *ne* to *nē*), sits close to Klaeber, so Wyatt against Klaeber is our two-editor reference. Over the whole text the editions probably agree more often than we measure: only 54 of a labelled sample of 100 disagreements between Wyatt's text and Harrison and Sharp's are editorial, the rest mostly apparent slips or a macron moved within a digraph. Wyatt and Klaeber agree on 96.08% of ambiguous tokens but only on 78.07% of class B. Both texts nearly always mark the pronouns *þē* 'thee' and *gē* 'ye' long, but in 89 of their 91 class B disagreements

Table 3: Agreement (%, κ of whole-token patterns in parentheses) on aligned *Beowulf* tokens (H&S, Wyatt: also 177 *Finnsburh* tokens). Lex.-amb.: lexicon-ambiguous. H&S: Harrison and Sharp.

	H&S, Wyatt	H&S, Klaeber	Wyatt, Klaeber
Aligned	16,833	16,577	16,918
All	97.73 (0.973)	98.82 (0.986)	97.67 (0.972)
Lex.-amb.	96.08 (0.942)	98.82 (0.983)	96.08 (0.942)
n	4,387	4,334	4,335
Class B	78.52 (0.469)	98.79 (0.974)	78.07 (0.460)
n	419	412	415

Table 4: Sampled error types (LLM labels, manually checked). Another edition agrees: the error matches another *Beowulf* edition.

	ByT5		CharsiuG2P	
Type	Prose	Verse	Prose	Verse
Gold not a candidate	3	0	0	0
Another edition agrees	0	12	0	5
Homograph (A)	9	48	26	29
Grammatical (B)	59	14	50	23
Editorial (C)	33	22	35	32
Items	104	96	111	89

Klaeber marks a *nē*, *gȳt* ‘yet’, *gē* ‘and’ or relative *þē* long and Wyatt’s text does not. On the 415 class B tokens that align in both, the prior agrees with Wyatt on 82.89% and with Klaeber on 71.08%, a swing of 11.81 points. There local2 gains +7.39 points over the word arm with Klaeber as gold, but only +2.41 [−0.40, +5.42] with Wyatt. Scored against Wyatt, local2 (85.30%) beats the two editors’ own agreement, which therefore does not cap accuracy.

The article *se* shows convention most clearly. Klaeber writes it short in 102 of 105 article tokens and long in every pronoun and relative token (by Brunetti’s glosses), while our training text mostly has *sē*. The prior gets none of the short articles right and no ByT5 run more than 14, and in line 2587, *þæt se mǣra maga Ecgðēowes*, the prior and every ByT5 run mark it long. Klaeber’s choice follows syntax and could be learned from Klaeber-style text, but a model cannot tell from bare text which convention the gold follows.

5.4 Errors, CharsiuG2P and limitations

Table 4 classifies 200 random ambiguous errors of the ByT5 sentence arm (seed 1, full cap) and 200 of CharsiuG2P (with Zupitza in the lexicon). ByT5’s prose errors are most often grammatical (*þe* 34 times) and its verse errors most often homographs (*Gēata* 10 times). Over A, B and C the two systems differ within prose (χ^2 test in SciPy [30], $p = 0.014$), where ByT5 makes fewer homograph errors, and within verse ($p = 0.013$), where it makes more. The labellers saw the word’s class for 270 of the 400 items, so label agreement is high partly by construction (κ 0.945 on the other 130).

CharsiuG2P. CharsiuG2P [37] is a multilingual ByT5 model that reads isolated words, trained on pronunciation lists (WikiPron’s for Old English). It was built for dictionary pronunciation, so our scores measure how well it fits our editors’ marking in running text. We run it as published (prefix `<ang>:`) and map its IPA to patterns with our aligner, counting unmappable outputs as wrong. We reproduce its published Old English score exactly (WER 0.208). Its evaluation stops decoding at Transformers’ default of 20 steps, truncating 12 of 500 outputs. With 50 steps the WER is 0.188, and about three quarters of word errors are length-only. On our text it gets length right on 70.96% of prose and 77.78% of verse tokens, well below the prior, and the context-free prior gets 87.09% of CharsiuG2P’s ambiguous prose length errors right from word habit alone.

Limitations. One test text per genre lets single words decide results (*Gēata*, *þe*), and most of the training text is Bright’s, as is the development text, so choices made on it may favour his conventions. The gold follows one editor, and we have not proofread the transcriptions.

6 Future work

If convention explains most *Beowulf* class B and *se* errors, training and testing in one editorial tradition should reduce them, and scoring against several editions would show what remains. The main caveats call for proofread test texts, ByT5 with all four sources, a dictionary-free tagger [11] for unlisted patterns (weaker than the best dictionary-based systems on unseen words), deterministic training [24] with five or more seeds, more verse and independent human labels for a sample.

References

[1] Anthropic. Claude Opus 5.5 (model id `claude-opus-5-5`), 2026. Used through Claude Code from 18 to 27 October 2026. URL: https://platform.claude.com/docs/en/models/opus-5-5/overview.

[2] J. W. Bright. *The Gospel of Saint John in West-Saxon*. D. C. Heath, Boston, 1904. With a glossary by L. M. Harris. Wikisource transcription. URL: https://en.wikisource.org/wiki/Index:Gospel_of_Saint_John_in_West-Saxon.djvu.

[3] J. W. Bright, editor. *The Gospel of Saint Matthew in West-Saxon*. D. C. Heath, Boston, 1910. Second, corrected printing (first printed 1904). Wikisource transcription. URL: https://en.wikisource.org/wiki/Evangelium_Secundum_Mattheum:_the_Gospel_of_Saint_Matthew_in_West-Saxon.

[4] J. W. Bright. *An Anglo-Saxon Reader*. Henry Holt, New York, third edition, 1913. Wikisource transcription. URL: https://en.wikisource.org/wiki/Bright%27s_Anglo-Saxon_Reader.

[5] J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. URL: https://doi.org/10.1177/001316446002000104.

[6] A. S. Cook and J. H. Pitman. *The Old English Physiologus*, volume LXIII of *Yale Studies in English*. Yale University Press, New Haven, 1921. Project Gutenberg ebook 14529. URL: https://www.gutenberg.org/ebooks/14529.

[7] E. Cullhed and A. Thörn Cleland. Stoicheia: Character-level masked diffusion for Ancient Greek textual restoration, parsing, and metrical scansion. arXiv:2608.07249, 2026. URL: https://arxiv.org/abs/2608.07249.

[8] J. Dodge, G. Ilharco, R. Schwartz, A. Farhadi, H. Hajishirzi, and N. A. Smith. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. arXiv:2002.06305, 2020. URL: https://arxiv.org/abs/2002.06305.

[9] B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, New York, 1993. URL: https://doi.org/10.1201/9780429246593.

[10] S. Esmail, K. Bar, and N. Dershowitz. How much does lookahead matter for disambiguation? Partial Arabic diacritization case study. *Computational Linguistics*, 48(4):1103–1123, 2022. URL: https://doi.org/10.1162/coli_a_00456.

[11] E. Gershuni and Y. Pinter. Restoring Hebrew diacritics without a dictionary. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 1010–1018, 2022. URL: https://doi.org/10.18653/v1/2022.findings-naacl.75.

[12] K. Gorman, G. Mazovetskiy, and V. Nikolaev. Improving homograph disambiguation with supervised machine learning. In *Proceedings of LREC 2018*, pages 1349–1352, Miyazaki, Japan, 2018. URL: https://aclanthology.org/L18-1215/.

[13] H. L. Hargrove, editor. *King Alfred's Old English Version of St. Augustine's Soliloquies*, volume XIII of *Yale Studies in English*. Henry Holt, New York, 1902. Wikisource transcription. URL: https://en.wikisource.org/wiki/King_Alfred%27s_Old_English_Version_of_St._Augustine%27s_Soliloquies.

[14] J. A. Harrison and R. Sharp, editors. *I. Bēowulf: An Anglo-Saxon Poem. II. The Fight at Finnsburh: A Fragment*. Ginn and Company, Boston, fourth, revised edition, 1894. First edition 1883. Project Gutenberg ebook 9701, in its revised version. URL: https://www.gutenberg.org/ebooks/9701.

[15] F. Klaeber, editor. *Beowulf and the Fight at Finnsburg*. D. C. Heath, Boston, 1922. Perseus Digital Library text Perseus:text:2003.01.0001. URL: https://www.perseus.tufts.edu/hopper/text?doc=Perseus:text:2003.01.0001.

[16] J. L. Lee, L. F. E. Ashby, M. E. Garza, Y. Lee-Sikka, S. Miller, A. Wong, A. D. McCarthy, and K. Gorman. Massively multilingual pronunciation mining with WikiPron. In *Proceedings of LREC 2020*, pages 4223–4228, 2020. Data: `ang_latn_broad.tsv`, retrieved 16 October 2026. URL: https://aclanthology.org/2020.lrec-1.521/.

[17] J. Martín Arista and D. Metola. UD Old English-OEDT: A Universal Dependencies treebank of Old English, 2025. Development branch. URL: https://github.com/UniversalDependencies/UD_Old_English-OEDT.

[18] T. Mokrowiecki. Acute accents as graphic markers of vowel quantity in two Late Old English manuscripts. *English Language and Linguistics*, 19(3):407–436, 2015. URL: https://doi.org/10.1017/S1360674315000015.

[19] N. Pangakis, S. Wolken, and N. Fasching. Automated annotation with generative AI requires validation. arXiv:2306.00176, 2023. URL: https://arxiv.org/abs/2306.00176.

[20] A. Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 8026–8037, 2019. URL: https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html.

[21] F. Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011. URL: https://jmlr.org/papers/v12/pedregosa11a.html.

[22] B. Plank. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of EMNLP 2022*, pages 10671–10682, 2022. URL: https://aclanthology.org/2022.emnlp-main.731/.

[23] Serg078. Conversion of the Clark Hall dictionary of Old English to TEI, 2026. Commit b53fc7d. Unlicense. Source: J. R. Clark Hall, *A Concise Anglo-Saxon Dictionary*, Project Gutenberg ebook 31543. URL: https://github.com/Serg078/Conversion-of-The-Clark-Hall-Dictionary-of-Old-English-to-TEI.

[24] C. Summers and M. J. Dinneen. Nondeterminism and instability in neural network optimization. In *Proceedings of ICML 2021*, volume 139 of *PMLR*, pages 9913–9922, 2021. URL: https://proceedings.mlr.press/v139/summers21a.html.

[25] H. Sweet. *An Anglo-Saxon Primer*. Clarendon Press, Oxford, eighth edition, 1905. Project Gutenberg ebook 34316. URL: https://www.gutenberg.org/ebooks/34316.

[26] J. Tauber. old-english-texts: `beowulf/raw-data/brunetti-length.txt`, 2021. Commit 1d944111. Brunetti's tokenisation and tagging of *Beowulf* with vowel lengths added from the Perseus text. CC BY-SA 4.0. URL: https://github.com/old-english-learner-texts/old-english-texts.

[27] A. Taylor, A. Warner, S. Pintzuk, and F. Beths. The York-Toronto-Helsinki parsed corpus of Old English prose (YCOE). University of York, distributed by the Oxford Text Archive, 2003. URL: https://varieng.helsinki.fi/CoRD/corpora/YCOE/.

[28] A. Thörn Cleland and E. Cullhed. Automatic annotation of Ancient Greek vowel length. arXiv:2608.01935, 2026. URL: https://arxiv.org/abs/2608.01935.

[29] O. Tichý, M. Roček, et al. Bosworth-Toller's Anglo-Saxon Dictionary online. LINDAT/CLARIAH-CZ, Charles University, 2021. Based on J. Bosworth and T. N. Toller (1898, Supplement 1921). CC BY 4.0. URL: http://hdl.handle.net/11234/1-3532.

[30] P. Virtanen et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. URL: https://doi.org/10.1038/s41592-019-0686-2.

[31] J. Winge. Automatic annotation of Latin vowel length. Bachelor's thesis in language technology, Uppsala University, 2015. URL: https://alatius.com/macronizer/winge2015.pdf.

[32] T. Wolf et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP 2020: System Demonstrations*, pages 38–45, 2020. URL: https://aclanthology.org/2020.emnlp-demos.6/.

[33] A. J. Wyatt, editor. *Beowulf*. Cambridge University Press, Cambridge, 1894. Wikisource transcription. URL: https://en.wikisource.org/wiki/Beowulf_(Wyatt).

[34] L. Xue, A. Barua, N. Constant, R. Al-Rfou, S. Narang, M. Kale, A. Roberts, and C. Raffel. ByT5: Towards a token-free future with pre-trained byte-to-byte models. *Transactions of the Association for Computational Linguistics*, 10:291–306, 2022. URL: https://doi.org/10.1162/tacl_a_00461.

[35] D. Yarowsky. Decision lists for lexical ambiguity resolution: Application to accent restoration in Spanish and French. In *Proceedings of ACL 1994*, pages 88–95, Las Cruces, New Mexico, 1994. URL: https://doi.org/10.3115/981732.981745.

[36] T. Ylonen. Wiktextract: Wiktionary as machine-readable structured data. In *Proceedings of LREC 2022*, pages 1317–1325, 2022. Data: https://kaikki.org/dictionary/Old%20English/, downloaded in October 2026. URL: https://aclanthology.org/2022.lrec-1.140/.

[37] J. Zhu, C. Zhang, and D. Jurgens. ByT5 model for massively multilingual grapheme-to-phoneme conversion. In *Proceedings of Interspeech 2022*, pages 446–450, 2022. Code and per-language results (commit 0c929390): https://github.com/lingjzhu/CharsiuG2P. URL: https://doi.org/10.21437/Interspeech.2022-538.

[38] J. Zupitza, editor. *Ælfrics Grammatik und Glossar. Erste Abteilung: Text und Varianten*, volume 1 of *Sammlung englischer Denkmäler in kritischen Ausgaben*. Weidmann, Berlin, 1880. Project Gutenberg ebook 45861. URL: https://www.gutenberg.org/ebooks/45861.

Use of AI tools

We used an AI assistant (Claude, Anthropic) for help with editing the code and checking the diction of this report, and to produce the labels described in Sections 3 and 5, which were then manually checked. We directed the work and take responsibility for its content.